

SUDARSHAN ANAND

AI/ML Researcher, Data Scientist

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EDUCATION

Georgia Institute of Technology

MS Computational Science and Engineering (CGPA: 4.0/4.0)

Atlanta, GA, United States

Aug 2024 – May 2026

Birla Institute of Technology and Science (BITS Pilani)

MSc. Mathematics and B.E. Computer Science (CGPA: 9.4/10)

Pilani, Rajasthan, India

Aug 2019 – Aug 2024

EXPERIENCE

Rezolve.ai - AI Product Development Intern

Jun 2025 – Aug 2025

- Engineered an enterprise **Retrieval-Augmented Generation (RAG)** search MVP using **FastAPI, Pinecone, and PostgreSQL**, integrating conversational memory to reduce retrieval to **<20-second latency**.
- Optimized platform UI/UX for employee onboarding by developing a **prefix-tree look-ahead search** for instant autocomplete suggestions, automating global trend pipeline updates via **GCP Cloud Run**.
- Architected an **Agentic AI ChatOps system** (Slack/Teams) to automate the IT incident response lifecycle, streamlining **alert triaging** and **dynamic runbook generation**, attracting pilot interest from leading **ITSM providers**.

Qure.ai Technologies Pvt. Ltd. - AI Scientist Intern

Jan 2024 – Jun 2024

- Optimized a **computer vision lung nodule detection** product, boosting AI-to-radiologist correlation by **~45%** to ensure strict clinical reliability for early cancer diagnosis.

PROJECTS (visit [here](#) for more information)

Multimodal Progression Tracking of Neurodegenerative disorders

Aug 2025 – present

- Architecting a multimodal **Concept Bottleneck Model (CBM)** to track Parkinson's progression, fine-tuning foundational models on **3,000+ 3D MRI scans** via **mixed-precision training** to guarantee highly explainable outputs.
- Scaling **distributed PyTorch pipelines** across **HPC environments using SLURM**, to process massive longitudinal patient cohorts, significantly reducing processing and model training times for complex temporal datasets.

Investigating importance of Patient Metadata in Public Health

May 2025 – present

- Spearheading a cross-functional study as the sole technical lead**, engineering end-to-end Python pipelines (pandas) to **extract signal from 1.2M unstructured patient records** while collaborating with subject-matter experts to translate clinical domain knowledge into quantitative logic.
- Independently drove the statistical analysis and visualization of high-dimensional datasets (Seaborn, Plotly)**, uncovering a **critical 94% data sparsity rate** that redefined the project's scope and heavily influenced downstream modeling assumptions.
- Prototyped and optimized GenAI entity-extraction pipelines using the Hugging Face API**; evaluated Gemma-2 via zero-shot and Supervised Fine-Tuning (SFT) to establish a baseline that quantified the impact of degraded metadata on LLM performance. Presented findings at the IEEE-EMBS BHI 2025 conference ([abstract](#)).

Samay: Time-series Foundational Models Library

Jan 2025 – May 2025

- Developed Samay, an open-source Python library** unifying 10 state-of-the-art foundational models, developing robust ML inference pipelines for time-series forecasting, classification, and anomaly detection.
- Implemented zero-shot and custom model fine-tuning workflows in PyTorch 2.0+**, successfully adapting pre-trained models to diverse data formats and specific tasks
- Evaluated framework performance** through rigorous quantitative analysis against the [GiftEval benchmark](#), validating the scalability of all 10 integrated models to match industry standards.

HONORS & AWARDS

IEEE BHI 2025 Data Challenge Competition Champion

Oct 2025

- Led the study design and development of **AI/ML-driven depression risk prediction**, winning the IEEE-sponsored competition (backed by NSF & Google). Particularly appreciated for deriving clinical insights from the model predictions.

IEEE BHI 2025 Young Professional NextGen Scholar (NSF, EMBS, Google Sponsored)

Sep 2025

- Awarded prestigious NSF/Google-sponsored scholarship for high-impact Biomedical AI research, including an invitation to present at the IEEE BHI 2025 conference.

PUBLICATIONS (visit [here](#) for full list)

- S. Anand, M. Lee, and S. S. Vasan**, 'Metadata makes Big Data useful for AI/ML-ready, equitable pandemic response', in IEEE-EMBS International Conference on Biomedical and Health Informatics 2025 1-Page Abstracts, 2025. (poster presentation)

SKILLS

Technical Skills: PyTorch, Pandas, Scikit-learn, Numpy, SQL, PostgreSQL, Pinecone, Git, SSH, Google Cloud Platform Slurm (HPC), Pydantic, Pytest

Machine Learning & Engineering: Foundational Models (LLMs), Model Fine-tuning (SFT, RLHF, DPO), Data Science, Large unstructured datasets, Scalable Inference, Vector Databases, Predictive Modeling, A/B testing.

Soft Skills: Technical writing (documentation, blogs), Leadership & Team management, Presentation, Collaboration